Quiz 5

Graded

Student

Clarisse Cheung

Total Points

0.5 / 2 pts

Question 1

Question 1

0 / 1 pt

– **0 pts** Correct

– **0.1 pts** Mostly Correct

– **0.5 pts** Half Correct

✔ – **1 pt** Not Correct

✔ – **0 pts** Defines a predictor that inverts A

– **0 pts** Does not define a predictor

1 Your writing is very faint, try to make sure your writing is better to read

2 Are you defining $negl(n)$ or claiming it exists due to being negligable

Question 2

Question 2

0.5 / 1 pt

– **0 pts** Full Credit

– **0.1 pts** Majority Credit

✔ – **0.5 pts** Half Credit

– **1 pt** No Credit

3 Theres a couple problems here:
1. You need to sample a uniformly from $\mathbb{Z}_N^*$ ($a \xleftarrow{\$} \mathbb{Z}_N^*$). This is different than choosing a arbitrarily.
2. We want $y' = y^* \cdot a^2$. Under your current implementation a correct computation for A would set $\hat{x} = y^* a$ but we're trying to calculate the root of y^*
3. If you do steps 1 and 2, the last step is to return $\hat{x} \cdot a^{-1}$.

CMSC 28400 **Quiz 5 (2 questions)** October 27, 2026

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Please write neatly, not too small, and not too faint.
Question 1 is one direction of Exercise 4.1. Question 2 is drawn from Exercise 4.3, except we have changed the definition of y^* to ensure there is no trivial solution, and asked only for pseudocode, no justification.

Question 1: Fix a randomized, polynomial-time algorithn Samp. On input 1^n, it outputs a pair (z, b), where $b \in \{0,1\}$. A **predictor** $\mathcal{A}$ is a PPT algorithm that receives z and outputs some $\hat{b} \in \{0,1\}$. For any predictor $\mathcal{A}$, define

$$f_{\mathcal{A}}(n) := \Pr_{(z,b)\leftarrow \mathsf{Samp}(1^n)}[\mathcal{A}(1^n, z) = b], \tag{1}$$

where the probability is over both the randomness of Samp and of $\mathcal{A}$. Suppose that for all predictors $\mathcal{A}$ there exists a negligible function negl such that, for all sufficiently large n,

$$f_{\mathcal{A}}(n) < \frac{1}{2} + \mathsf{negl}(n). \tag{2}$$

Prove that for all predictors $\mathcal{A}$, the function

$$\left| f_{\mathcal{A}}(n) - \frac{1}{2} \right| \tag{3}$$

is negligible.

1 Assume (2). We must prove an upper and lower bound.

2 We build a PPT func $\bar{A}$:

$\bar{A}(1^n, z)$:
  $b' = A(1^n, z)$
  return $1 - b'$

$\bar{A}$ is PPT since it calls A once with a subtraction

3. $\bar{A}$ is right exactly when A is wrong. If $b' = b$, then $1 - b \neq b'$ and thus $\bar{A}$ is wrong. If $b \neq b'$, then $1 - b' = b$ and $1 - b = b'$ thus $\bar{A}$ is right.

4. Define $\mathsf{negl}(n) = f_A(n) - 2^{-n}$. Since there exists a negligible function negl, and a threshold n_1, we have $f_A(n) < \frac{1}{2} + \mathsf{negl}(n) < f_A(n) + 2^{-n}$

5.

Question 2: Define some deterministic function Special such that for every $N \geq 2$, $y^* := \mathsf{Special}(N)$ is such that $y^* \in QR_N$. In other words, Special picks out some particular "special" quadratic residue y^*. Suppose $\mathcal{A}$ is a PPT principle-root finding algorithm such that

$$\Pr\left[(\widehat{x}^2 \equiv y \mod N) \wedge (\widehat{x} \in QR_N)\right] = \varepsilon(n) \tag{4}$$

in the probability experiment

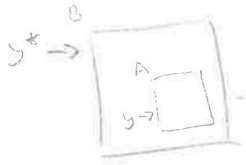
$$N \leftarrow \mathsf{Blum}(1^n) \tag{5}$$
$$y \xleftarrow{\$} QR_N \tag{6}$$
$$\widehat{x} \leftarrow \mathcal{A}(N, y). \tag{7}$$

Give the pseudocode for a PPT algorithm $\mathcal{B}$ such that

$$\Pr\left[(\widehat{x}^2 \equiv y^* \mod N) \wedge (\widehat{x} \in QR_N)\right] = \varepsilon(n) \tag{8}$$

in the probability experiment

$$N \leftarrow \mathsf{Blum}(1^n) \tag{9}$$
$$y^* := \mathsf{Special}(N) \tag{10}$$
$$\widehat{x} \leftarrow \mathcal{B}(N, y^*). \tag{11}$$



B(N, y*):
some $a \in \mathbb{Z}_N^*$
$y' = (y^*) \cdot a^2$ or $y' = y^* \cdot a^2$
$\hat{x}^* := A(N, y')$
return